

Computer Architectures
Exam of 15/12/2022

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First name, Last name, ID:

Question 1 (4 points)

The Tomasulo architecture for superscalar processors with dynamic scheduling and speculation often uses a Reorder Buffer (ROB)

You are requested to

1. Explain what the ROB is and how it works
2. Describe its architecture
3. Detail when data are written in the ROB, when an entry is marked as busy and when it is released
4. List the advantages stemming from the use of the ROB.

Write your answer here.

[illegible]

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[illegible]

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Question 3 (6 points)

An 8 x 8 matrix BOARD of bytes stores A-Z ascii characters only, inclusive of two and only two occurrences of the letter “A”. Write an 8086 assembly program which determines if the two “A”s are on the same column. Please observe/comply with the following

- It is mandatory to cut the matrix by rows.
- In your solution, please provide the declaration of BOARD and the code, together with a short description of the algorithm used and significant comments to the code and instructions.
- It is guaranteed that the BOARD only stores A-Z ascii characters and that there are two and only two occurrences of the character “A”.
- As this is an assembly program, please do NOT design an algorithm which is suitable to a high-level language approach, but strongly focus on the cut by rows of the matrix and its related properties.
- ANY (EVEN PARTIAL) BRUTE FORCE APPROACH IS NOT ACCEPTABLE.
- Please observe that no “solution” will be shared/discussed by professors. Students are invited to discuss their solutions in their study/social groups.
- Hint: to devise a suitable algorithm, take as an example a smaller matrix (e.g. 4x4), “write it” when cut by rows, and identify the property of elements on the same column.

Examples:

Matrix BOARD (with “A”s on the same column)

C	D	A	F	K	K	J	M
B	B	B	D	H	G	R	E
O	O	P	U	Y	R	E	F
W	W	W	W	F	R	Y	Z
T	T	T	T	T	T	T	T
D	E	A	H	T	U	I	O
R	E	R	T	S	W	E	T
B	T	U	O	K	Z	X	D

Matrix BOARD (with “A”s not on the same column)

C	D	F	F	K	K	J	M
B	B	B	D	H	A	R	E
O	O	P	U	Y	R	E	F
W	W	W	W	F	R	Y	Z
T	T	T	T	T	T	T	T
D	E	A	H	T	U	I	O
R	E	R	T	S	W	E	T
B	T	U	O	K	Z	X	D

Write your code in a file saved in the 8086 folder.

Click on the following link to open a web page with the 8086 instruction set:
<http://www.jegerlehner.ch/intel/IntelCodeTable.pdf>

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Question 4 (9 points)

The IEEE-754 SP standard expresses floating-point numbers in 32 bits:

31	30	23	22	0
sign	exponent		mantissa	

Write the `addFPpositiveNumbers` subroutine, which receives in input two 32-bit numbers, considers them as IEEE-754 SP floating point numbers, and returns their sum (in the same format). Bit 31 of the two input numbers is always 0 (i.e., the two numbers are positive).

In details, the subroutine implements the following steps:

1. take the mantissa of the two parameters
2. set bit 23 of the mantissa to 1
3. compare the two exponents. If they are equal, the exponent of the result is the same. If they are different:
 - a) the exponent of the result is the highest one
 - b) shift right the mantissa of the number with the lower exponent by as many position as the difference between the two exponents.
4. sum the two mantissas: this is the mantissa of the result. If bit 24 of this mantissa is 1:
 - a) shift right the mantissa of the result by one position
 - b) increment the exponent of the result by one.
5. set bit 23 of the mantissa of the result to 0.
6. combine the mantissa and the exponent to get the final result.

Example:

```
parameter1 = 0100 0010 0100 1011 0000 0000 0000 0000 = 424B0000
parameter2 = 0100 0001 1010 0100 0000 0000 0000 0000 = 41A40000
mantissa1 = 0000 0000 0100 1011 0000 0000 0000 0000 = 004B0000
mantissa2 = 0000 0000 0010 0100 0000 0000 0000 0000 = 00240000
mantissa1 = 0000 0000 1100 1011 0000 0000 0000 0000 = 00CB0000
mantissa2 = 0000 0000 1010 0100 0000 0000 0000 0000 = 00A40000
exponent1 = 1000 0100 = 84
exponent2 = 1000 0011 = 83
exponentResult = 1000 0100 = 84
mantissa2 = 0000 0000 0101 0010 0000 0000 0000 0000 = 00520000
mantissaResult = 0000 0001 0001 1101 0000 0000 0000 0000 = 011D0000
mantissaResult = 0000 0000 1000 1110 1000 0000 0000 0000 = 008E8000
exponentResult = 1000 0101 = 85
mantissaResult = 0000 0000 0000 1110 1000 0000 0000 0000 = 000E8000
result = 0100 0010 1000 1110 1000 0000 0000 0000 = 428E8000
```

Important notes:

1. Create a new project with Keil inside the "ARM" directory and write your code there. You can add the subdirectories "led" and "button" to your project if you need them.
2. The assembly subroutine must comply with the ARM Architecture Procedure Call Standard (AAPCS) standard (about parameter passing, returned value, callee-saved registers).
3. Click on the following links to open web pages with the ARM instruction set

<http://www.keil.com/support/man/docs/armasm>

<https://developer.arm.com/documentation/ddi0337/e/Introduction/Instruction-set-summary?lang=en>

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Question 5 (4 points)

Add a C file (e.g. sample.c) to the project created in the previous exercise.

Write here the main function (which needs to be called from the Reset handler).

Inside the main function, call the `addFPpositiveNumbers` subroutine, passing two floating-point numbers. If the result is higher than 3.1415, you have to switch on led 4 and switch off all other leds. Otherwise, switch on led 5 and switch off all other leds.